Ayush Rathod

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Summary

Full Stack Engineer with a backend-heavy foundation, specializing in production-grade AI systems. Shipped complete products across React/Next.js frontends, FastAPI and Node.js backends, and async task infrastructure, with hands-on depth in RAG pipelines, multi-agent orchestration, and LLM evaluation. Experienced with PostgreSQL, Kafka, Redis, Docker, and AWS.

Work Experience

DevDynamics • Software Engineering Intern Jul 2025 – May 2026
Pune, India (On-site)

- Built a Python **LLM-as-judge pipeline** to score AI code review tools against **100 human-curated PRs** across 5 open-source repositories, measuring **precision and recall** to enable version-over-version quality regression tracking and direct comparison against competitor tools.
- Developed an **AST-based symbol-resolution pipeline** (Tree-sitter, 19 languages) that extracts identifiers from changed diff lines and resolves them to precise cross-file definitions, providing LLMs with impact-scoped context for code review, raising precision **12% → 54%** and recall **39% → 71%**.
- Reduced peak AWS Lambda memory from **4.5 GB → under 700 MB** and codebase processing time **2 min → 30 sec** by designing a two-tier AST cache — an in-memory LRU layer that stores only serialized `DocumentIR` (releasing native tree-sitter objects immediately after extraction) backed by a PostgreSQL persistence layer that survives cold starts.
- Shipped a **MCP server** deployed on AWS ECS that exposes DevDynamics engineering analytics like DORA metrics, PR cycle times, ticket status, and contributor profiles, has **OAuth 2.1-secured, LLM-callable tools**, enabling AI agents to query live team performance data in natural language across multi-tenant engineering organizations.
- Integrated **Claude Code, Cursor, Windsurf, and GitHub Copilot** APIs into DevDynamics' AI Impact & Adoption module for per-user acceptance rates, session activity, and PR contribution metrics; and **Jira** into Refacto.ai's review pipeline to validate code changes against ticket requirements and acceptance criteria.

Projects

Advista – AI Competitive Intelligence Platform Demo GitHub
FastAPI, LangChain, LangGraph, Groq, Celery, Redis, React, PostgreSQL, Prisma, Docker, AWS Lambda, Firebase

- Built a **competitive-intelligence platform** using a **LangGraph** multi-turn chatbot that schema-enforces LLM extraction of user intent into an 8-field Pydantic `ResearchBrief`, fans out **5 concurrent SerpAPI queries** across Google Search, Google Forums, and YouTube (with transcript extraction via `YouTubeTranscriptApi`), and synthesizes results into a **7-section market-intelligence report**, persisting full session state through a 5-stage status machine in **PostgreSQL via Prisma**.
- Separated long-running **SerpAPI** search tasks from the HTTP request cycle using **Celery + Redis** for task dispatch, result polling, and retry fault tolerance with a feature-flagged `asyncio.gather` fallback for AWS Lambda (Docker container image), and **Firebase Admin SDK** for user lifecycle management (creation, email verification, password reset) synchronized to a PostgreSQL-backed **JWT** auth layer.

ValueX – Wealth Management AI Microservice Demo GitHub
FastAPI, OpenAI, Python, Server-Sent Events, scikit-learn, Pydantic, yfinance, Docker, pytest

- Engineered a **multi-agent AI pipeline** from scratch with a **single-call structured intent classifier** routing across 8 specialist agents, a **pre-LLM safety guard** completing in under 10ms, and **real-time streaming via SSE** all under a **$0.05/query cost ceiling** at gpt-4.1 pricing with p95 first-token latency under 2s.
- Built a **labeled eval suite** against 100+ labeled queries achieving $\geq$**85% routing accuracy** and $\geq$**95% safety recall**, with full **CI coverage via mocked LLM** and a Portfolio Health agent producing structured risk outputs (concentration, benchmark alpha, actionable observations) across edge-case user profiles.

Education

Vishwakarma Institute of Information Technology Jun 2022 - May 2026
Bachelor of Technology in Artificial Intelligence and Data Science *CGPA: 8.59*

Achievements

Grand Finalist • Smart India Hackathon 2024: Top 5 nationally (AICTE, Ministry of Law & Justice)

5x Hackathon Winner: Recognized for developing cloud-native and AI-driven solutions in various hackathons

Technical Skills

Languages: JavaScript, TypeScript, Python, C++
Tech: React.js, Next.js, Node.js, FastAPI, GraphQL, REST APIs
AI/ML: LLM Integration, Prompt Engineering, Context Engineering, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Model Context Protocol (MCP), Transformers, LLM Evaluation, Vector Databases.
Data: PostgreSQL, MongoDB, Redis, Apache Kafka
Cloud/DevOps: AWS, GCP (Compute Engine), Docker, GitHub Actions CI/CD, Linux, Bash